

Xiangyu Zhong (钟翔宇)

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RESEARCH INTERESTS

My research lies at the intersection of **deep learning**, **wireless communication**, and **optimization**. I am focused on advancing **edge intelligence** with **connectivity**, **robustness**, **adaptivity**, and **security**. I am also passionate about score-based methods in **generative models** and **multimodal large models**, and their long-horizon scalable **edge deployment** in the future.

EDUCATION

The Chinese University of Hong Kong (CUHK)

Department of Information Engineering

Ph.D., Information Engineering (*with CUHK Vice-Chancellor's PhD Scholarship*)

Supervisor: Ying-Jun Angela Zhang, IEEE Fellow

Aug. 2023 - Jun. 2027

Hong Kong SAR

University of Electronic Science and Technology of China (UESTC)

School of Information and Communication Engineering (SICE)

(Honghu Elite Project, National Key Laboratory on Wireless Communications, UESTC)

B.Eng. (*with Honors*), Communication Engineering

GPA: 3.99/4.00, Rank: 1/230

Sept. 2019 - Jun. 2023

Chengdu, China

PUBLICATION *corresponding author

- **Xiangyu Zhong**, Xiaojun Yuan, Ying-Jun Angela Zhang, "FedCova: Robust Federated Covariance Learning Against Noisy Labels", *Under Review, Advances in Neural Information Processing Systems (NeurIPS)* 2025.
- Wenjing Yan, **Xiangyu Zhong***, Xiaolu Wang, Ying-Jun Angela Zhang, "Momentum-Driven Adaptivity: Towards Tuning-Free Asynchronous Federated Learning", *Forty-second International Conference on Machine Learning (ICML)* 2025.
- **Xiangyu Zhong**, Chenxi Zhong, Xiaojun Yuan, Ying-Jun Angela Zhang, "UAV Enabled Over-the-Air Federated Learning: A Hierarchical Aggregation Approach", *Under Revision, IEEE Transactions on Wireless Communications*.
- **Xiangyu Zhong**, Xiaojun Yuan, Huiyuan Yang, Chenxi Zhong, "UAV-Assisted Hierarchical Aggregation for Over-the-Air Federated Learning", *IEEE Global Communications Conference (GLOBECOM)* 2022, pp. 807–812.

AWARDS

- **Outstanding Students Awards**, CUHK Dec. 2024
- **Second Prize**, Four Minute Thesis Competition, IEEE Communications Society June. 2024
- **Talent Development Scholarship**, HKSAR Government June. 2024
- **CUHK Vice-Chancellor's PhD Scholarship** Feb. 2023
- **The Most Outstanding Students Award of UESTC** (10 of graduates each year) Dec. 2022
- **National Scholarship** (2 times) Oct. 2020 & Oct. 2022
- Tang Lixin Scholarship (60 awardees over 40000+ undergraduates, postgraduates, and Ph.D.) Jan. 2021
- Scholarship of Gratitude for Chinese Modern Scientists (10 students per year) Nov. 2022
- Shiqiang First Class Scholarship (Ranking 1st of 6 winners from juniors and seniors) Oct. 2021
- **Outstanding Winner** (0.08% globally), American Mathematical Contest in Modeling (MCM)/ Interdisciplinary Contest in Modeling (ICM) Apr. 2021
- **AMS Award** (0.014% globally), American Mathematical Society, American MCM/ICM Apr. 2021
- **National Honorable Mention**, National Mobile Communication 5G Technology Competition Jun. 2021

- **First Prize**, Sichuan Province, National Mobile Communication 5G Technology Competition Jun. 2021
- First Class Scholarship for Outstanding Students of UESTC (3 times) Sept. 2020 & Sept. 2021 & Sept. 2022
- **Silver Prize**, Sichuan Province, *Internet+* Innovation & Entrepreneurship Competition Oct. 2021
- Third Prize, Sichuan Province, *Challenge Cup* Science & Technology Academic Competition May 2021
- **Outstanding Graduate of Sichuan Province** Nov. 2022
- **First Prize and 4A Grade** (highest level), National Oral English Test Contest for College Students Jun. 2021
- Grade-A Comprehensive Quality Certificate, Sichuan Province (Quickest to acquire in history) Jan. 2021
- **Honorary Title of Liren Elite** (5 students awarded for outstanding leadership) Dec. 2020
- Outstanding Volunteer of the University Enrollment Aug. 2020 & Jul. 2022
- Research and Competition Talent Award of *Glory SICE* Dec. 2021

LEADERSHIP & INFLUENCE

- Secretariat Member, UESTC Hong Kong Alumni Association Sep. 2024 - Present
- Founder and President, Graceland Music Club, CUHK Oct. 2023 - Sep. 2024
- General Administrator, Postgraduate Hall Residents' Association, CUHK Oct. 2023 - Sep. 2024
- President, Sunshine Volunteer Association, UESTC Jul. 2021 - Jul. 2022
- Class President, Liren Leadership Class, UESTC Sept. 2020 - Mar. 2022
- Minister, Department of Art and Sport, Students' Union, UESTC Jun. 2020 - Jun. 2021
- Peer Counselor, Student Development Center, UESTC Sept. 2022 - Jun. 2023
- Tutor, Peer Support Association, SICE, UESTC Sept. 2021 - Jun. 2023
- Student Tutor, Grade 2022, Glasgow College, UESTC Jul. 2022 - Jun. 2023
- Singer, Electric-Acoustic Band, Collegiate Art Troupe, UESTC Sep. 2019 - Jan. 2022

PROJECT EXPERIENCES

Intelligent Spectrum Sensing & Innovative Wireless Communication Jun. 2020 - Jun. 2022
Honghu Elite Project, National Key Laboratory on Wireless Communications, UESTC Chengdu, China
 Supervisor: Prof. Ying Liu, Senior Member, IEEE

- Learned core courses one semester earlier than the normal schedule but with a more challenging level, such as Principle of Communication(93/100), Information Theory(90/100), and Statistic Signal Analysis(90/100).
- Completed the *Honghu Elite Level Up Project* from *Primary Level*, *Middle Level*, *Advanced Level*, to *Innovative Level* with the evaluation score of 91/100, 93/100, 95/100, and 97/100 respectively, **ranking 1st in the project**.
- Proposed a creative correction algorithm utilizing the unaligned pseudo-frame structure in the project *Dual Channel Interference Correction and Phase Compensation*, leading to the best performance with an extra 2dB gain.

Modeling and Application for 3-D Vector Assessment System May 2021 - Feb. 2022
School of Mathematical Sciences, UESTC Chengdu, China
 Supervisor: Prof. Zhiyong Wang, Member, Expert Panel, CUMCM & Party Secretary, UESTC Affiliated Education Group

- Won the **Outstanding Winner** (0.08% globally, 6 best papers of Problem E) and **AMS Award** (0.014% globally, only 1 among the 6) in MCM/ICM, with the latter being awarded by the American Mathematical Society.
- Completed the competition season as the chief of paper writing in English by Latex, the crucial contributor to modeling in 3DVA and CST models, core inspector of algorithms and simulations, and the team leader.
- Won the First Prize (1.6%) of the Simulation Competition in UESTC of MCM/ICM.
- Broke the history of UESTC to be the first team with sophomores only awarded as the Outstanding Winner.
- Been invited to publish the winning paper of MCM/ICM in COMAP, **as the chief author and team leader**.
- Proposed a Three-dimensional Vector Assessment Model (3DVA) for a visual and comprehensive evaluation, and a Curved-Surface Target Model (CST) for the target of development set and dynamical realization. **The models and ideas were cited by later research teams in 2022 MCM/ICM.**